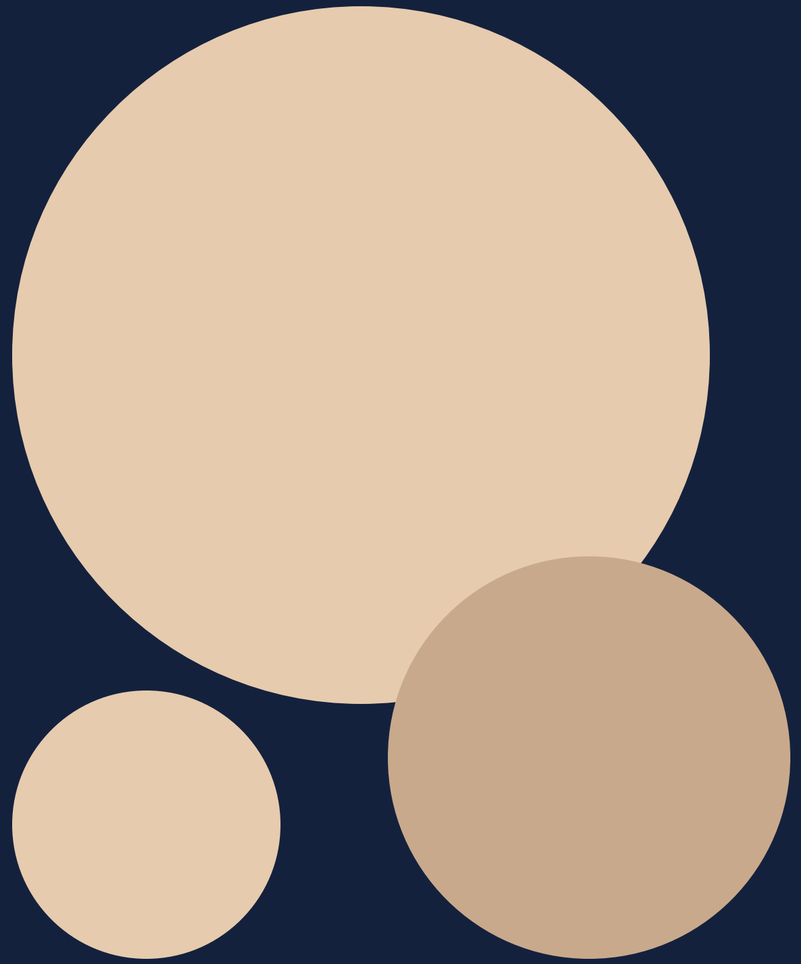
MAAI 902 · Assessment 2

Teaching a computer to read procurement paperwork

AI-Powered Procurement and Spare Parts Document Intelligence System



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github.com/halakhalid8991/MAAI902_Assessment2_NLP_Solution_Design_Hala_Sarhan

THE PROBLEM

Too many documents, not enough hours

Hundreds a day
Purchase orders, supplier emails, maintenance reports and general letters.

Reading by hand is costly
Wrong filing, missed reorders, duplicate orders and late decisions.

ERP can't read free text
It handles fields well, but the meaning sits in the wording.

From: site supervisor · Subject: stock alert

URGENT: gearbox GB 300 stock at the Jebel Ali Free Zone site is critically low, please raise an emergency PO

One email, four facts, found automatically

- **Part · gearbox GB 300**
- **Site · Jebel Ali Free Zone**
- **Urgency · critical**
- **Action · emergency PO**

THE SOLUTION

Six steps from raw text to answers

1 · Data
550 synthetic documents in four types

2 · Preprocessing
Clean the text, then turn it into TF-IDF vectors and embeddings

3 · Classification
A Linear SVM sorts each document by type

4 · Key details
Entities, urgency and sentiment, read from the original text

5 · Ask and answer
Hybrid search, a refusal gate and Qwen2.5-3B on the GPU

6 · Evaluation
Scored against gold labels, with unseen-template and noisy tests

INPUT DATA

550 documents, built to be tested hard

DOCUMENT TYPE	COUNT
Purchase order	150
Supplier email	150
Maintenance report	150
General letter	100

Unseen templates
Train on templates 0 to 6, test on 7 and 8

Messy scans
137 flagged (about 25%), 110 actually altered by OCR-style typos

Tricky emails
About 15% of supplier emails mix good and bad news

Generated in Python with a fixed seed (42), so every run gives the same data. Real procurement documents are confidential, so I made my own and ran 17 automated checks on it (16 passed).

PREPROCESSING

Three light cleaning steps

1

Tidy the text

Lowercase, collapse spaces, drop odd symbols. Numbers stay, so gb 300 survives.

2

Spell out shortcuts

PO becomes purchase order, ETA becomes estimated time of arrival.

3

Drop filler words

Remove words like the, is and of, but keep the domain terms.

Entities and sentiment skip all of this and read the original text, because capital letters and exact spelling carry signal, like URGENT.

REPRESENTATIONS

Two ways to turn words into numbers

FOR CLASSIFICATION

TF-IDF vectors

5,000 features, single words and word pairs, fitted on training data only.

- Small data: just 550 documents
- Fast and predictable
- Weights can be inspected for audit

FOR SEARCH

Dense embeddings

all-MiniLM-L6-v2, 384 numbers per document, kept in a FAISS index and encoded on the GPU in 1.7 seconds.

- Matches meaning, not exact words
- “Shipment is late” finds “delivery delayed”
- Small enough for a free T4

Tested: five classifiers, and keyword vs meaning vs hybrid search. Not tested: character n-grams, BM25 and bigger embedding models.

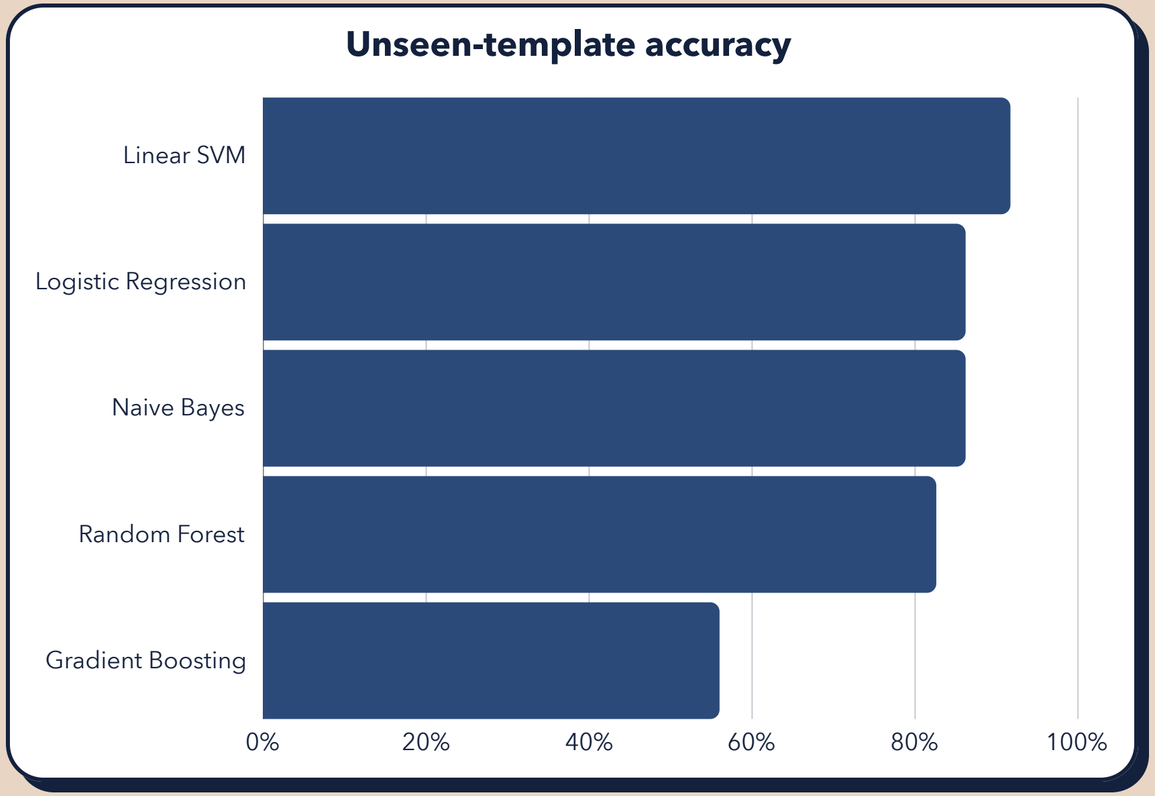
RESULTS · CLASSIFICATION

100% sounds great. 91.7% is the real number

91.7%

Linear SVM accuracy on unseen templates

Cross-validation mostly rewards memorising templates, and nearly every model scores 99 to 100% there.
Weak spot: general letters, recall 0.56 (8 of 18 missed, 7 of them filed as supplier emails).



RESULTS · ENTITY EXTRACTION

Sharp on clean text, shaky on noisy text

1.000
F1, clean

0.916
F1, noisy

0.853
Recall, noisy

Regular expressions find dates and quantities. Lookup lists find known suppliers, sites, equipment and parts. Precision stays high (0.989), but exact matching misses damaged names like PartsPro Internat1onal.

RECALL ON NOISY TEXT, BY ENTITY

Date 0.974 · Quantity 0.940
Part 0.872 · Equipment 0.822
Site 0.782 · Supplier 0.779
Next step: fuzzy matching

RESULTS · URGENCY AND SENTIMENT

Simple rules, honest limits

URGENCY · PURCHASE ORDERS

100%

Routine, high or critical, from cue words like “critical”, “urgent” and “no rush”. Scored on 150 clean purchase orders and 39 noisy ones.
The templates spell the cues out, so don't expect this on real documents.

SENTIMENT · SUPPLIER EMAILS

86.0%

Positive, negative or neutral by counting phrases. 150 emails, 21 errors: 14 negative called positive, 7 neutral called negative.
Average confidence is 0.91, higher than the accuracy deserves.

QUESTION ANSWERING

Answers that stay tied to the documents

1 · Retrieve
Keyword, meaning and hybrid search over all 550 documents

→

2 · Refusal gate
Best similarity under 0.53? Skip the model and say no

→

3 · Generate
Qwen2.5-3B reads the top five documents and cites them

→

4 · Measure
In testing, answer entities must appear in the retrieved documents, and citations must be real

Why bother?

Language models can say fluent things that no source supports. The gate, a prompt that demands citations and a fixed fallback (“Information not found in the available documents”) limit that risk, and the evaluation measures how well they work.

RESULTS · SEARCH AND ANSWERS

No retriever wins everything

RETRIEVER	HIT@1	HIT@3	MRR	P@5
TF-IDF	0.600	0.667	0.662	0.433
Dense	0.433	0.800	0.597	0.340
Hybrid (RRF)	**0.567**	**0.767**	**0.681**	**0.440**

30 queries. TF-IDF wins on exact part codes (MRR 1.000 on maintenance). Dense wins on reworded supplier emails (0.733 vs 0.173). Hybrid has the best MRR, but only just.

15/15
off-topic and unanswerable questions refused

0%
real questions wrongly refused

100%
entity grounding in answers

86.7%
citation precision (purchase orders 0.767)

RESULTS · OVERALL

Putting it all together



PART	SCORE	WEIGHT
Classification, unseen templates	**91.7%**	25%
Entities, noisy F1	**91.6%**	25%
Sentiment accuracy	**86.0%**	10%
Retrieval, hybrid Hit@3	**76.7%**	20%
Answers, citation precision	**86.7%**	20%

The weights are my own judgement. This is a solid base for a pilot on real data, not a measure of real-world accuracy.

ETHICS

Four risks and how I handle them

01 Privacy

UAE PDPL. Synthetic data only for now. In production: mask names, limit purpose, log access.

02 Language bias

English only today, so Arabic documents could be mishandled. Fix: multilingual models.

03 Supplier fairness

Models may favour documents like the training data. Monitor accuracy by supplier type.

04 Made-up answers

Citations, a refusal gate and a fixed fallback; grounding is measured in testing. People approve orders and payments.

LIVE DEMONSTRATION

Let's see it working

TAB 1 · Analyse a document
Type, entities, urgency, sentiment and similar documents

TAB 2 · Batch upload
.txt, .eml and .csv files in, one results table out

TAB 3 · Ask the documents
Answers with citations, or a polite refusal

Try this one: “Who won the football world cup?” and watch it refuse.

Open the Colab notebook

See the GitHub repo

LIMITS AND NEXT STEPS

Feasible, not yet proven on real documents

What it can't do yet

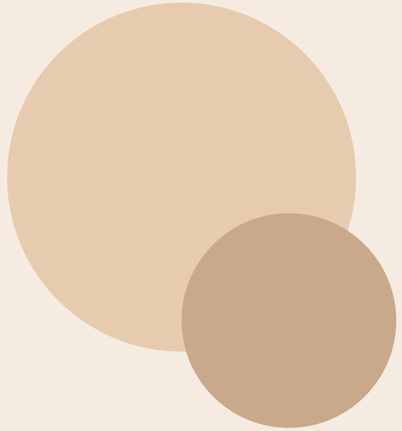
- Synthetic, template-based data
- Small test sets; gate tuned on the same queries
- Urgency and sentiment are keyword rules
- Entities match known names only
- English only; always picks one of four types

What I'd do next

- Pilot on real, anonymised documents
- Fuzzy matching for damaged names
- Fine-tuned NER and pre-trained sentiment
- Arabic and English
- Falcon LLM for UAE data sovereignty

SUMMARY

It reads, extracts and answers, or says it can't



91.7% on unseen templates

0.916 entity F1 on noisy text

15 of 15 bad questions refused

Thank you! Questions?